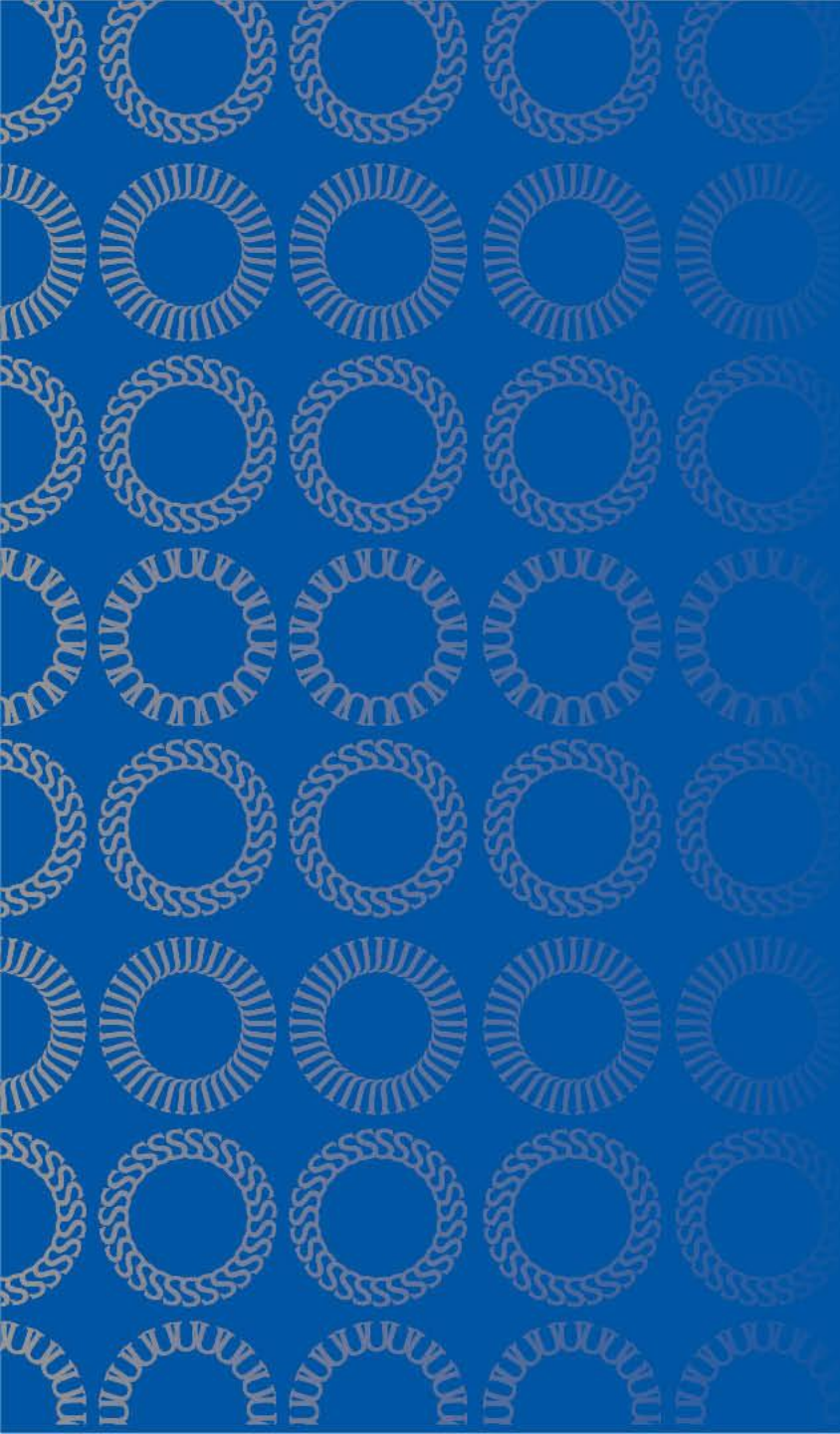




DATA 255 Deep Learning Technologies

Nov 13, 2025

Taehee Jeong, Ph.D.

Generative AI for images

- Autoencoder / VAE
- Generative Adversarial Networks (GAN)
- Diffusion model

What are the key components (blocks) of Autoencoder?

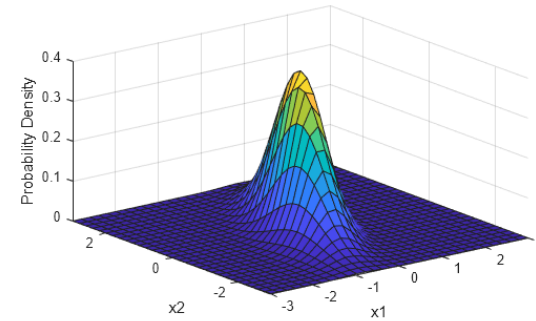
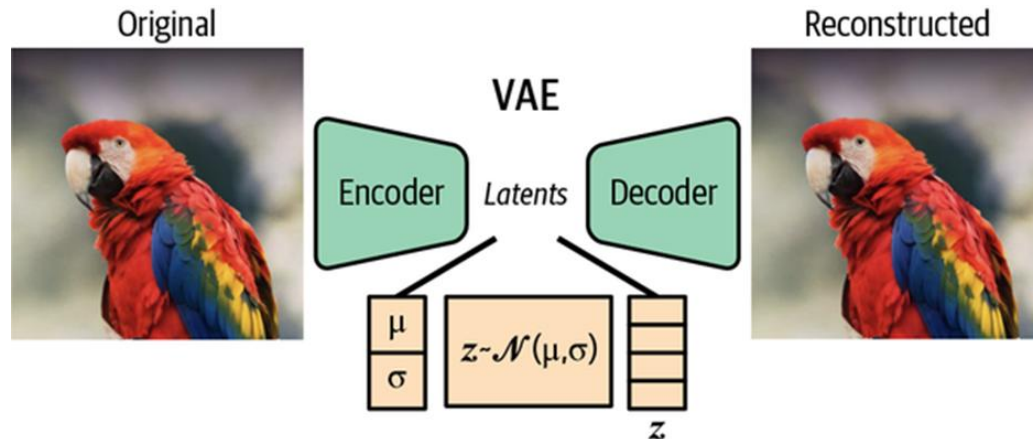
- Down-sampling using Encoder
- Up-sampling using Decoder

What are the key components of Decoder block?

- Transpose convolution (or de-convolution)
- Skip connection
- Concatenation

What is the difference between autoencoder and Variational autoencoder?

- In autoencoder, each image is mapped directly to one point in the latent space.
- In VAE, each image is mapped to a multivariate normal distribution around a point in the latent space



$$\frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

What are the two networks of GAN?

- Discriminator
- Generator

What are the two networks of GAN?

- Discriminator → binary image classification
- Generator → autoencoder

How to train network in Autoencoder and Generator in GAN?

- autoencoder → MSE loss in pixel level
- Generator in GAN → loss of discriminator

Why Diffusion model?

It generates higher-quality images than those produced by GANs.

What are the two processes for Diffusion model?

- forward diffusion process
- reverse diffusion process

Forward diffusion process

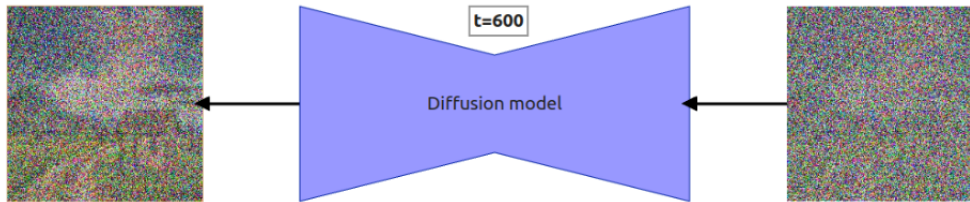
- Gradually adds Gaussian noise to the data
- *It systematically and slowly destroy structure in a data distribution*

$$x_t \approx \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon$$

$\epsilon \sim N(0,1)$

Reverse diffusion process

- It predicts the entire noise to be removed with a given timestep

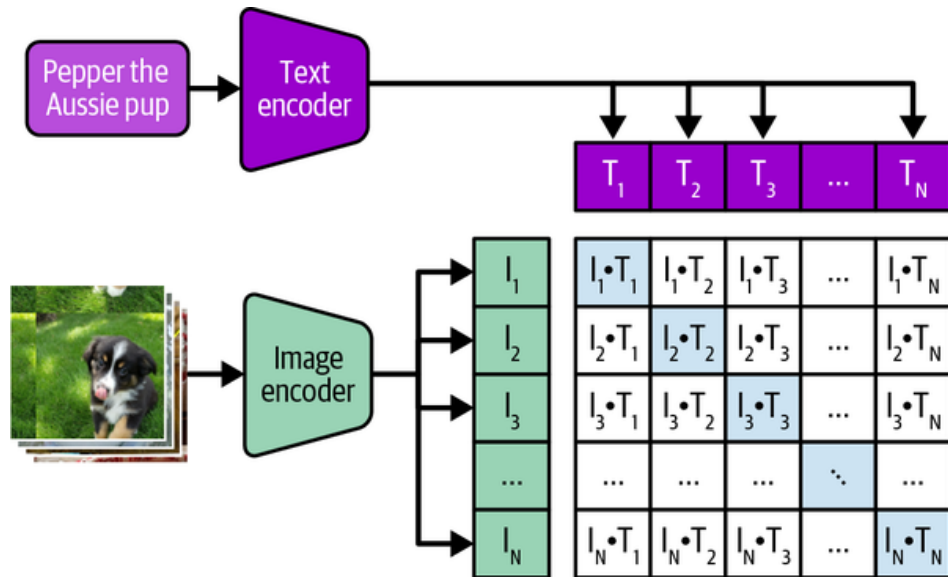


$$x_{t-1} \approx \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(x_t, t) \right) + \sigma_t z$$

$z \sim N(0,1)$

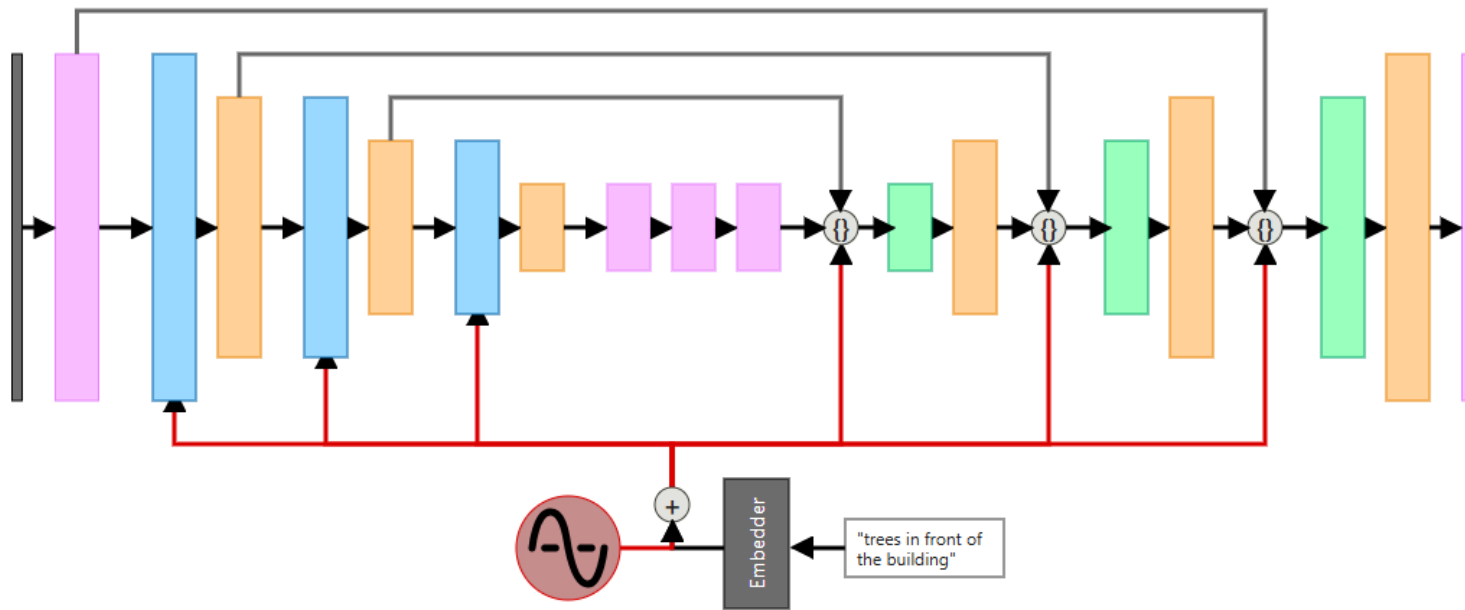
Text-image generation using diffusion process

- Implement text-embedding into diffusion process using Clip/text-encoder



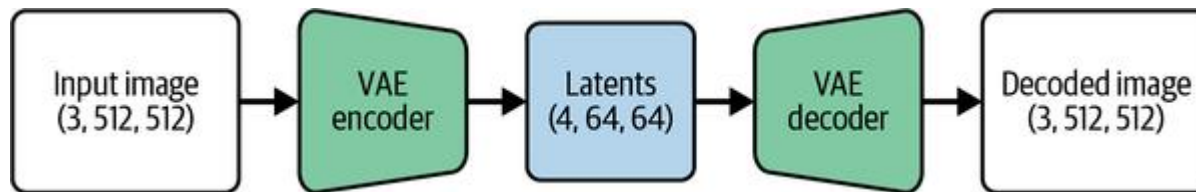
Text-image generation using diffusion process

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What is stable diffusion (or latent diffusion)?

- As image size grows, so does the computational power required to work with those images.
- Using smaller images, we can get benefits for lower memory usage, fewer layers needed in the UNet, faster generation times.



Grading criteria: Project

Criterion	Description	Points
1. Project Domain / Problem	Clearly explain the area or domain your project belongs to (e.g., medical imaging, scene understanding, anomaly detection, etc.). Describe the problems /challenges in this domain you are targeting.	2.0
2. Current Baseline	Describe the existing works (what baseline deep learning models exist). What is the current status or limitations of existing works, and what gaps are still remained.	2.0
3. Dataset and Evaluation Metrics	Identify the relevant dataset(s) used in prior works and specify the evaluation metric(s) (e.g., accuracy, FID) and baseline performance.	2.0
4. Your Approach / Technical Novelty	Explain your approach in terms of technical novelty or architectural improvement compared to the baselines (existing works from your literature survey), and how it aims to overcome the identified limitations.	2.0
5. Results	What are the performance of your approach? Compare it with the existing works (baselines) using the evaluation metrics.	2.0

Group project presentation schedule

- Nov 13 : 2, 1, 6, 7, 9
- Nov 20 : 3, 4, 5, 8